

IR PCB

0) Powering (From Power PCB)

Power PCB (5V) → Teensy VIN

Power PCB (GND) → Teensy GND

Power PCB (3.3V) → IR PCB 3.3V rail

Power PCB (GND) → IR PCB GND rail

Entry bulk capacitor (your single cap)

- 100 μ F / 25V from 3.3V → GND at the 3.3V entry
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1) Power to every TSSP58038 (from Teensy 3.3V)

Teensy 3.3V → IR PCB 3.3V rail (Vin for all TSSP58038)

Teensy GND → IR PCB GND rail

Oled Screen VCC → IR PCB 3.3V rail

Oled Screen GND → IR PCB GND rail

For each sensor:

- TSSP58038 Vin (pin 3) → IR PCB 3.3V rail
 - TSSP58038 GND (pin 2) → IR PCB GND rail
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2) Sensor outputs (D0–D15 nodes) → Teensy analog pins

Each channel is (already on your PCB):

IRoutX → 10k → Dx, and Dx → 0.22 μ F → GND

Wire Dx to Teensy 4.1:

- D0 → A0
- D1 → A1
- D2 → A2

- D3 → A3
 - D4 → A4
 - D5 → A5
 - D6 → A6
 - D7 → A7
 - D8 → A8
 - D9 → A9
 - D10 → A10
 - D11 → A11
 - D12 → A12
 - D13 → A13
 - D14 → A14
 - D15 → A15
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3) UART to the other Teensy (ball direction link) — Serial2 (TX2/RX2)

Using Serial2 on **Sensor Teensy4.1**

- TX2 (pin 8) → Main Teensy RX
- RX2 (pin 7) ← Main Teensy TX

Camera to IR PCB Teensy :

RX_C → Pin 34

TX_C → Pin 35

RST → 1K ohm resistor → Pin 36